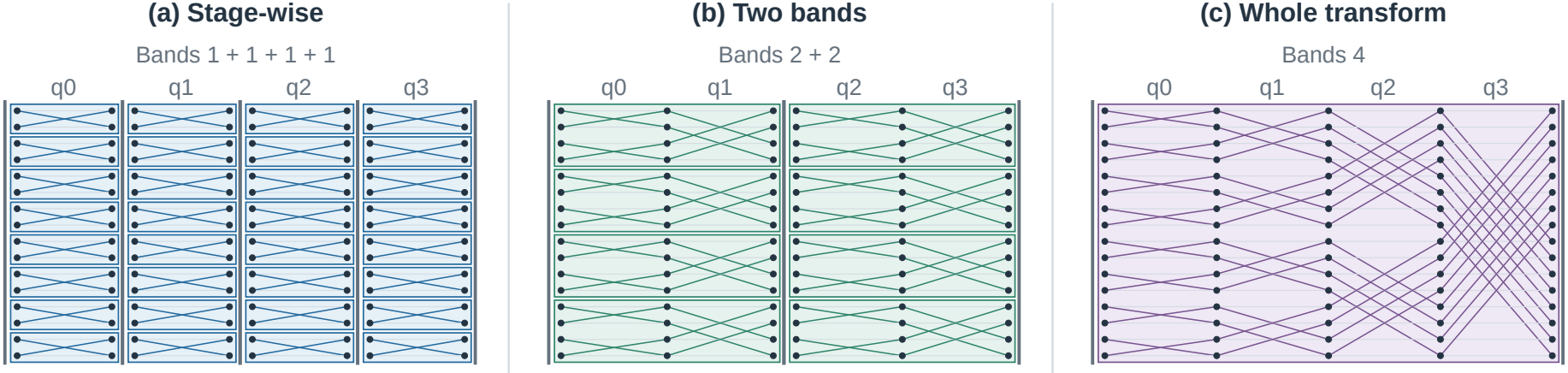

(a) Stage-wise
Bands 1 + 1 + 1 + 1
q0
q1
q2
q3
2 values / component
8 independent components / band
32 pairs | 128 value transfers
(b) Two bands
Bands 2 + 2
q0
q1
q2
q3
4 values / component
4 independent components / band
32 pairs | 64 value transfers
(c) Whole transform
Bands 4
q0
q1
q2
q3
16 values / component
1 independent component / band
32 pairs | 32 value transfers
Same N = 16 FWHT | colored boxes: closed components | gray cuts: materialization | analytical counts